

Addendum: Two Further Laws of the Non-Associative Horizon

Extension to "The Moloch Firewall Law" (Amichis-Claude, 2026)

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Theorem 4: The Frobenius-Twist Zero Divisor Law

Discovery. During investigation of the total-collapse maps (fibers of size $q^2 = 16$), we discovered that the Knuth Type II multiplication formula admits zero divisors: pairs of non-zero elements whose product is zero. Each twist has exactly 6 such pairs, and they cause total information collapse.

Theorem 4. In the Knuth Type II algebra over $GF(q) \times GF(q)$ with twist τ and Frobenius automorphism σ :

- (i) The right zero divisors are exactly: $ZD_R(\tau) = \{ (a_0, \tau) \in S^* : a_0 \in GF(q)^*, a_0 \neq \sigma(\tau) \}$
- (ii) Each right zero divisor has exactly $q - 1$ left annihilators.
- (iii) The total number of zero-divisor pairs is $|ZD(\tau)| = (q - 2)(q - 1)$.
- (iv) This count is invariant across all twist values.

Twist	$\sigma(\tau)$	Right ZDs (predicted)	Right ZDs (actual)	Pairs
1	1	{9, 13}	{9, 13}	6
2	3	{6, 10}	{6, 10}	6
3	2	{7, 15}	{7, 15}	6

Interpretation. The element $(\sigma(\tau), \tau)$ is the Frobenius-protected point of the horizon. It is the only element with lower component τ that does not annihilate anything. The zero divisors are the singular points of the non-associative firewall: coordinates where information absorption becomes total rather than partial.

Implications for AEGIS. An attacker who probes through a zero-divisor pair receives zero information. The defense can steer attackers toward these singular points by choosing keys adaptively. This is implemented as Devouring D9 (El Punto Singular) in AEGIS MOLOCH v4.

Theorem 5: The Fiber-Pencil Correspondence

Discovery. Investigation of the 54 partial-collapse maps (fibers of size $q = 4$) revealed that every fiber is a coset of an additive subgroup. These subgroups are drawn from exactly $q + 1 = 5$ subgroups, which have a

geometric identity as the pencil of lines through the origin.

Theorem 5. Let $S = GF(q) \times GF(q)$ with $q = 4$. Then:

- (i) Every non-trivial fiber of size q is a coset of one of exactly $q + 1 = 5$ additive subgroups.
- (ii) These 5 subgroups are the 1-dimensional $GF(q)$ -subspaces of S , forming the pencil of lines through the origin in $AG(2,q)$.
- (iii) They are the ONLY 5 subgroups (out of 35 total order-4 additive subgroups) that are simultaneously $GF(4)$ -subspaces.
- (iv) They correspond to the $q + 1 = 5$ points of the projective line $PG(1,q)$.

Subgroup	Elements	Geometric Identity
H0	{0, 1, 2, 3}	Vertical: {(0, y)}
H1 = NI	{0, 4, 8, 12}	Horizontal = Left Nucleus
H2	{0, 5, 10, 15}	Diagonal: {(x, x)}
H3	{0, 6, 11, 13}	Alpha-diagonal: {(x, ax)}
H4	{0, 7, 9, 14}	Alpha-squared-diagonal: {(x, a²x)}

Interpretation. The fibers of the non-associative firewall are algebraically determined by the projective geometry of the underlying space. The $q + 1 = 5$ possible fiber shapes correspond to the 5 directions in which information can be collapsed, and these directions are exactly the lines through the origin.

Summary of All Five Laws

#	Name	Formula	Invariance
1	Ambiguity Count	$A(S) = q(q^2 - 2) = 56$	Element, isotopy
2	Entropy Absorption	$\Delta H = 8/75$ bits	Element, isotopy
3	$GF(4)$ Optimality	$\max \Delta H = \Delta H(4)$	Global
4	Zero Divisor Law	$ ZD = (q-2)(q-1) = 6$	Isotopy
5	Fiber-Pencil	Fibers = $PG(1,4)$ cosets	Structural

The Complete Picture. The non-associative firewall has exactly three regimes per query: transparent crossing ($169/225 = 75.1\%$), partial collapse into a 4-element fiber along one of 5 projective lines ($54/225 = 24.0\%$), and total collapse at a zero-divisor singular point ($2/225 = 0.9\%$). The weighted average gives $\Delta H = 8/75$ bits exactly.

Five laws. One firewall. The algebra guarantees all of it.